

[T]-Theory: Neuroscience

P5 · Conscious Hardware · For: The Neuroscientist

The Cortical EMF Propagator — from synapse to macroscopic CEMI field.

Scale 5–7 identification:

- Scale 5: Synaptic impulse response — axonal propagation kernel
- Scale 6: Cortical EMF propagator — CEMI field from synchronised firing
- Scale 7: Conscious percept pole — phase transition at threshold T_c

BRECVEMA as neural basis:

- **Brainstem reflex** (scale 5) — automatic survival response
- **Rhythmic entrainment** (scale 6) — temporal synchronisation
- **Emotional contagion** (scale 6–7) — mirror neuron field
- **Cognitive appraisal** (scale 7) — prefrontal modulation

I · The Cortical EMF Propagator

At scale 6, the USF propagator identifies with McFadden's CEMI field:

$$G_{\text{CEMI}}(\mathbf{r}, \tau) = \frac{1}{4\pi\epsilon_0} \sum_i \frac{q_i \delta(\tau - r_i/c)}{r_i}$$

The synchronised firing of pyramidal neurons generates a macroscopic electromagnetic field with coherence length $\ell = 1/k_6$. As $k_6 \rightarrow 0$ at T_c , global integration onset = consciousness.

II · The 8D BRECVEMA Field

The emotional field lives in 8-dimensional BRECVEMA mechanism space:

$$\Psi = \sum_{\alpha=0}^7 \psi_{\alpha} \hat{e}_{\alpha}$$

where $\hat{e}_{\alpha} \in \{\text{BS, RE, EC, CO, VI, EM, ME, AJ}\}$. The coupling matrix $W_8 \in \text{Mat}_{8 \times 8}(\mathbb{R})$ has eigenvalues = propagator poles = normal modes of emotional processing.

Largest eigenvalue ≈ 2.0 — the Nostalgia attractor pole (P11).

III · Neural Implementation

- **BS**: amygdala · startle circuit · basal ganglia
- **RE**: motor cortex · basal ganglia · cerebellum
- **EC**: mirror neuron network · insula · ACC
- **CO**: limbic system · hippocampus · parahippocampal cortex
- **VI**: visual cortex · DMN · semantic memory
- **EM**: prefrontal · DMN · DLPFC
- **ME**: orbitofrontal · VMPFC · memory consolidation
- **AJ**: integrated evaluative scalar — gauge mode

IV · Experimental Signatures

- EEG γ -band coherence $\propto k_6^{-1}$ during emotional episodes
- fMRI BOLD signal = time-averaged $|\Psi|^2$ projection
- TMS as volitional injection $J_{\text{user}}(t)$ — direct field perturbation
- Heart rate variability as τ_m proxy (memory kernel decay constant)

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